

Narendiran Gopinathan Chembu

cgnarendiran@gmail.com +1 (669) 699 8598 **Staff Machine Learning Engineer** | San Jose, US

EDUCATION

University of Amsterdam (UvA)
Masters in **Artificial Intelligence**
2018-2020,
Amsterdam, The Netherlands

**Indian Institute of Technology
Madras (IITM)**
B. Tech in **Mechanical Engineering**
(Minor: **Industrial Engineering**)
2013-2017, Chennai, India

LINKS



FRAMEWORKS

PyTorch, HuggingFace
Transformers, Ray, OpenCV, ROS /
ROS2, CUDA

LANGUAGES

Python, JavaScript, C++

INFRA

Docker, Git, AWS / GCP

COURSES

LLM / VLM: Fine-tuning (LoRA, SFT),
RAG & vector databases, agentic
workflows (ReAct), batch inference
& prompt caching, LLM-as-judge
evaluation, causal & counterfactual
benchmarking

PUBLICATIONS

CausalDriveBench: A Causal
Reasoning Benchmark for Driving
Vision-Language-Action Models -
NeurIPS 2026 (under review), with
Renesas and IIT Delhi

Journal RSC Aqueous Dispersions
of Lipid Nanoparticles Wet
Hydrophobic and
Superhydrophobic Surfaces, 2017,
Soft Matter, Royal Society of
Chemistry

WORK EXPERIENCE

Fastcode.AI | **Staff ML Engineer**

Aug' 2023 - Present (3 years) | Bangalore, India -> San Jose, US

- Leading ADAS world-model engagement with **Renesas**, a 6-month program building an action-conditioned driving world model for R-Car edge silicon.
- Authored CausalDriveBench, a **NeurIPS** submission with **Renesas** and **IIT Delhi**; causal reasoning benchmark for driving VLAs built on causal scene graphs over nuScenes; generated 7.3k structured visual QA pairs and 1k alternative trajectories across Pearl's Causal Hierarchy, and evaluated 13 models to expose near-chance causal accuracy and no correlation between language and trajectory performance.
- Built an AI legal research solution for **Miai Law** with vector-based retrieval over **500k+** Australian cases; enabled case research, contract audits, and statute search via indexed crawls, delivered as a full-stack B2C web app with auth and file handling.
- Spearheaded the development of AutoQRA - an AI-driven quantitative risk analysis system for **Saudi Aramco** that automated the digitization of complex PFDs/P&IDs using custom DETR models, advanced OCR, and graph-based analysis, reducing QRA preparation time by **85%** and achieving detection accuracies above **98%**.
- Led the Vital Sensing project at Mercedes-Benz R&D - contactless heart-rate estimation from in-cabin NIR video; built the signal extraction and denoising pipeline achieving **<5 BPM MAE**, **89%** accuracy under low-light and motion artifacts, slated for upcoming production vehicle systems.

Minds.AI Technologies | **ML Engineer**

Mar' 2022 - Jul' 2023 (1 year 4 months) | Bangalore, India

- Trained an RL setup switching agent (in SMT2020 fab testbed) that reduced queue lengths of wafer lots by **50%**, leading to significant improvements in cycle times and throughput; these results attracted new customers including GlobalFoundries and Intel

NewSpace Research and Technologies | **Robotics and ML Engineer**

Apr' 2021 - Mar' 2022 (1 year) | Bangalore, India

- Developed a fixed-wing flight stack MVP in under three months: offboard PX4 control (SITL, HITL, in-flight), Gazebo/Unity3D flight-dynamics simulation, decentralized collision avoidance (RVO2), a ROS2-ROS1 bridge, and a PyQt5 ground control station for mission planning and geofencing.

CBoost | **Robotics and ML Engineer**

Oct' 2020 - Mar' 2021 (6 months) | Breda, The Netherlands

- Built an autonomous field robot (Jetson AGX, NVIDIA Isaac SDK) from scratch in two months with custom stereo visual odometry, AprilTag relocalization, and obstacle avoidance; trained SegNet + HoughCNet in tandem for a production crop-row detection pipeline at 0.87 IoU.

ZyLAB | **ML Research Engineer** (Thesis project)

Nov' 2019 - Aug' 2020 (10 months) | Amsterdam, The Netherlands

- Determined the most efficient loss-centric method for unsupervised domain adaptation of BERT for entity recognition (**+0.26 F1**); contributed a pre-processed Enron email dataset and annotation set.

Center For Innovation (student-run org. in IITM) | **Software Team, Abhiyaan**

Aug' 2016 - Aug' 2017 (1 year) | Chennai, India

- Institute autonomous ground vehicle team - placed 13th of 34 global teams at IGVC 2017 (Michigan, USA); implemented a custom pure-pursuit planner for sharp-turn differential drive and EKF pose estimation fusing wheel encoders, GPS, and IMU.